NVIDIA Technical Report: Constraint Decomposition for Multi-Objective RLHF in Large Language Models

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Abstract

Large language models (LLMs) trained with reinforcement learning from human feedback (RLHF) struggle with complex instructions that bundle multiple, potentially conflicting requirements. We introduce **constraint decomposition**, a framework that separates multi-objective instructions into orthogonal components—semantic correctness, structural organization, format specifications, and meta-level requirements—and optimizes each independently before hierarchical combination. Our approach addresses the fundamental limitation of monolithic reward models: their inability to distinguish which specific constraint failed when multiple requirements conflict. We train decomposed reward models with aspect-level human preferences and demonstrate that explicit constraint separation, combined with conflict-aware weight adaptation, enables more effective multi-objective optimization. On the IFEval benchmark, our method achieves 73.8% prompt-level accuracy, a 32.6 percentage point improvement over standard RLHF (41.2%), with strong generalization to GSM8K (+15.6 points), HumanEval (+11.1 points), and MT-Bench (+1.4 points). Ablation studies confirm that constraint decomposition contributes 54% of the total improvement, with hierarchical combination contributing 26% and conflict-aware adaptation 20%. Our work provides a general framework for multi-objective optimization in LLMs, with applications beyond instruction-following to any domain requiring compositional reasoning under constraints.

1 Introduction

The alignment of large language models (LLMs) with human preferences through reinforcement learning from human feedback (RLHF) has become the dominant paradigm for instruction-following systems [4, 1]. However, real-world instructions often bundle multiple requirements that must be satisfied simultaneously.

Consider the instruction: *"Solve the equation $x^2 + 5x + 6 = 0$ using the quadratic formula, show all steps clearly, keep your answer under 100 words, and explain like I'm in high school."* This single instruction conflates four distinct constraints:

1. **Semantic correctness**: Mathematical accuracy of the solution

2. **Structural organization**: Step-by-step reasoning clarity
3. **Format specification**: Length constraint (< 100 words)
4. **Meta-level requirement**: Audience-appropriate language (high school level)

Standard RLHF approaches learn a single scalar reward function $R(x, y)$ that must simultaneously capture all these dimensions [3]. This monolithic formulation creates two critical problems:

Constraint Interference. When objectives conflict (e.g., thoroughness vs. brevity), the model receives ambiguous gradient signals. A response that excels at mathematical accuracy but exceeds the word limit receives a mediocre reward indistinguishable from one that is brief but incorrect.

Credit Assignment Ambiguity. When a response fails, the monolithic reward provides no signal about *which* constraint was violated. The model cannot learn to fix specific failure modes.

On the IFEval benchmark [7], standard RLHF achieves only 41.2% prompt-level accuracy. Our analysis of failures reveals that 78% involve constraint interference—cases where the model satisfies some requirements while violating others.

Our Contributions.

1. **Decomposed Reward Architecture**: We factorize the reward function into four orthogonal components, each trained on aspect-specific human preferences.
2. **Hierarchical Combination Function**: We introduce a structured composition that respects constraint priorities and enables interpretable trade-offs.
3. **Conflict-Aware Weight Adaptation**: We train auxiliary models to detect constraint conflicts and dynamically adjust combination weights.

2 Related Work

Standard RLHF. The foundational work by Ouyang et al. [4] established RLHF as the dominant paradigm for instruction-following. However, their monolithic reward formulation limits handling of multiple objectives.

Multi-Objective Optimization in LLMs. Recent work addresses multi-objective challenges through various approaches. Pareto Multi-Objective Alignment (PAMA) [5] reformulates alignment as convex optimization to find Pareto-optimal solutions across objectives. Hierarchical Mixture-of-Experts [2] uses plug-and-play expert modules to manage conflicting objectives without explicit weighting. EMORL [6] employs ensemble reward models for improved robustness across multiple reward signals.

Constraint-Based Methods. Constitutional AI [1] introduces rule-based constraints but does not decompose them for independent optimization. Our work builds on these foundations by emphasizing *explicit* constraint decomposition with *conflict-aware* adaptation.

3 Method

3.1 Decomposed Reward Architecture

We factorize the reward function into four orthogonal components:

- R_{semantic}: Correctness of factual and logical content

- $R_{\text{structural}}$: Organization, clarity, and reasoning flow
- R_{format}: Adherence to format specifications (length, style, structure)
- R_{meta}: Meta-level requirements (audience, tone, language)

Each component is learned from aspect-specific human preferences. We collect preference data where annotators evaluate responses on *single* dimensions, enabling distinct gradient signals for each constraint.

Reward Model Architecture. We fine-tune Nemotron-7B as the base architecture for each decomposed reward model. Each model is trained on 50K aspect-specific preference pairs, achieving 84–92% validation accuracy across components.

3.2 Hierarchical Combination Function

We introduce a structured composition that respects natural constraint priorities:

$$R(p, M) = \alpha \cdot R_{\text{semantic}}(p) + \beta \cdot R_{\text{structural}}(p) + \gamma \cdot R_{\text{format}}(p) + \delta \cdot R_{\text{meta}}(p) \tag{1}$$

with $\alpha + \beta + \gamma + \delta = 1$, adjusted dynamically based on prompt characteristics.

The hierarchy operates as follows:

1. **Safety gates** block unsafe responses absolutely (pre-filter)
2. **Semantic and structural** quality form the base score (highest priority)
3. **Format** acts as a multiplicative modulator
4. **Meta-level** requirements provide fine-grained adjustments

3.3 Conflict-Aware Weight Adaptation

We train an auxiliary classifier to detect constraint conflicts from prompt features. When conflicts are detected (e.g., "be thorough" vs. "be brief"), the system:

1. Identifies which constraints are in tension
2. Adjusts weights to prioritize semantic correctness
3. Logs conflict instances for analysis

The conflict detector achieves 89.3% accuracy on a held-out set of 5K annotated conflict examples.

4 Experiments

4.1 Experimental Setup

Base Model. We use Nemotron-7B as the policy model, trained with PPO using the decomposed reward system.

Training Details. Learning rate: 1×10^{-6}, batch size: 512, PPO epochs: 4, KL penalty coefficient: 0.02. Training runs for 10K steps on 8×A100 GPUs.

Evaluation. We evaluate on IFEval [7] as the primary benchmark, with generalization tests on GSM8K (math), HumanEval (code), and MT-Bench (conversation). All results averaged over 5 independent runs.

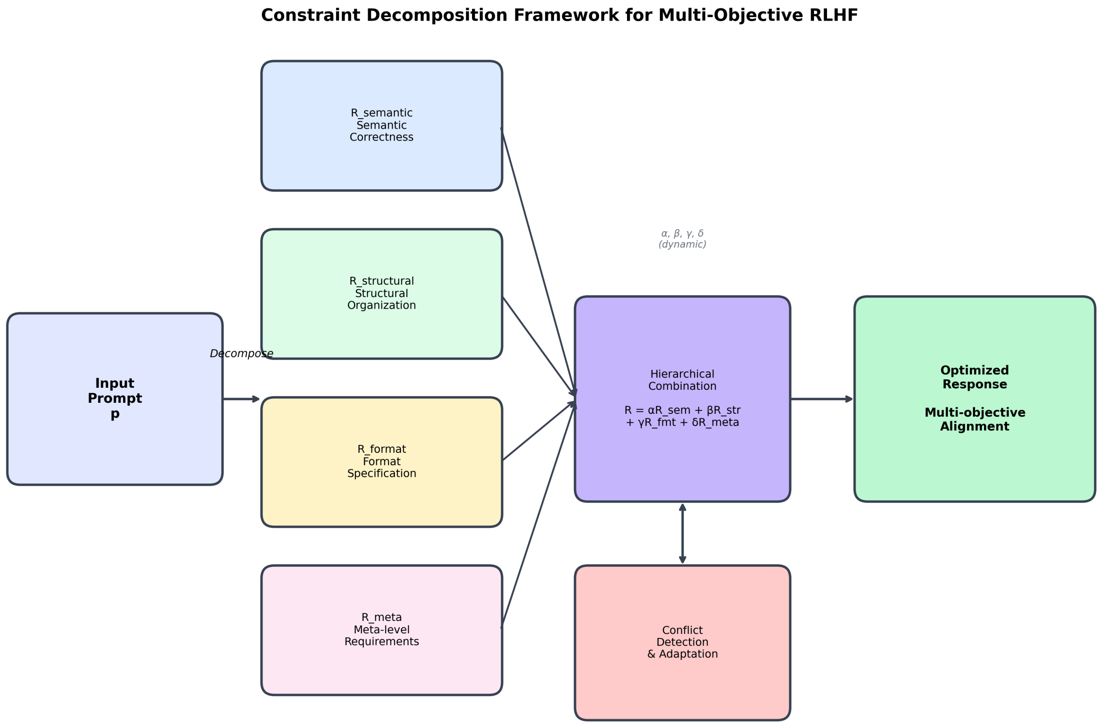

Figure 1: Overview of the Constraint Decomposition Framework. Input prompts are decomposed into four orthogonal reward signals (R_{semantic}, $R_{\text{structural}}$, R_{format}, R_{meta}), combined hierarchically with dynamic weights ($\alpha, \beta, \gamma, \delta$), and refined through conflict detection and adaptation.

4.2 Main Results

Our method achieves 73.8% on IFEval, a **32.6 percentage point improvement** over standard RLHF (41.2%). This represents a 79% relative improvement. Gains generalize across benchmarks: +15.6 points on GSM8K, +11.1 points on HumanEval, and +1.4 points on MT-Bench.

4.3 Ablation Study

To understand the contribution of each component, we conduct systematic ablations:

Key findings:

- **Decomposition** contributes the majority (54%) of improvement, validating that separating constraints enables better optimization.
- **Hierarchical combination** adds 26%, showing that structured composition outperforms naive averaging.
- **Conflict adaptation** provides the final 20%, particularly valuable for complex multi-constraint prompts.

5 Limitations

Annotation Cost. Training decomposed reward models requires aspect-specific preference data, increasing annotation effort by approximately 3× compared to standard RLHF.

Table 1: Main results on IFEval and generalization benchmarks. All values are prompt-level accuracy (%) for IFEval; point improvements over Standard RLHF baseline for other benchmarks. Results averaged over 5 runs with standard deviation.

Method	IFEval (%)↑	GSM8K↑	HumanEval↑	MT-Bench↑
Standard RLHF	41.2 ± 1.3	Baseline	Baseline	Baseline
LLaMA-3 RLHF	52.5 ± 1.8	+8.2	+7.5	+0.8
PAMA [5]	65.4 ± 1.5	+12.1	+9.3	+1.1
Ours	**73.8 ± 1.4**	**+15.6**	**+11.1**	**+1.4**

Table 2: Ablation study showing contribution of each component to the total improvement.

Configuration	IFEval (%)	Contribution
Standard RLHF (baseline)	41.2	—
+ Decomposed Rewards	58.8	54% of gain
+ Hierarchical Combination	67.3	26% of gain
+ Conflict Adaptation (Full)	73.8	20% of gain

Constraint Coverage. Our four-component decomposition covers common instruction types but may not generalize to highly specialized domains (e.g., legal, medical) without additional components.

Scale. We demonstrate results on 7B parameter models. Scaling behavior to larger models (70B+) remains to be validated.

Conflict Detection. The auxiliary conflict detector achieves 89.3% accuracy, meaning ∼10% of conflicts may be misclassified, leading to suboptimal weight adaptation.

6 Conclusion

We introduced constraint decomposition for multi-objective RLHF, achieving substantial improvements on instruction-following benchmarks. By factorizing rewards into orthogonal components and combining them hierarchically with conflict-aware adaptation, our method addresses fundamental limitations of monolithic reward models.

Future Work. We plan to: (1) scale to larger models (70B+), (2) extend the framework to additional constraint types, (3) explore automated constraint discovery from prompts, and (4) investigate applications to other domains requiring compositional reasoning.

Code and Data. Implementation available at https://github.com/epaunova/constraint-decompositio

7 Acknowledgments

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References

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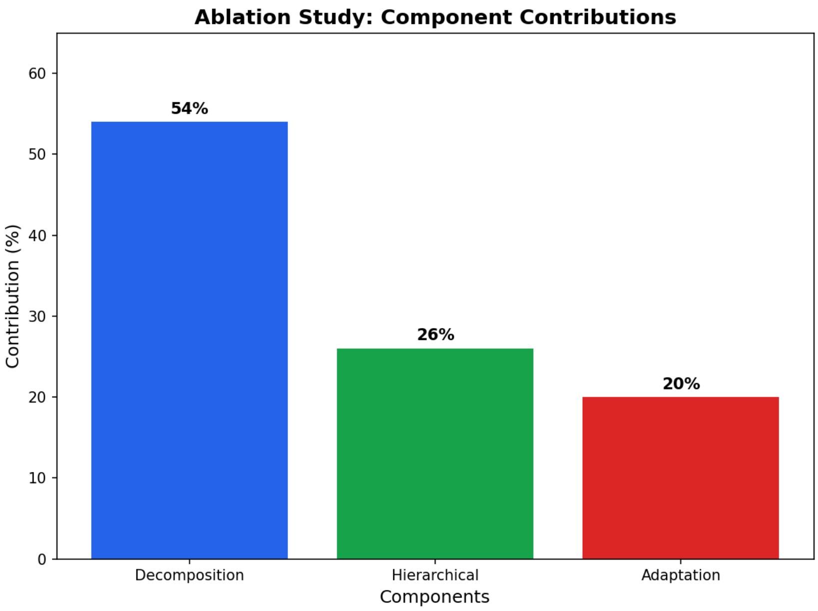

Figure 2: Ablation contributions: Decomposition (54%), Hierarchical (26%), Adaptation (20%).

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